Jacob Muniko

Technology Builder | Web, AI and Automation | Founder of Linklet

Karatina, Kenya · Available for remote work · Portfolio: https://muniko53.github.io/jacob-muniko-portfolio/

Status as of November 2026

Professional Summary

I am a technology builder who creates practical software, intelligent systems and automation that solve real problems. I start from a problem, design a workable solution, build it, and improve it through testing and iteration. My work spans web applications, desktop software, AI-assisted tools and workflow automation, and I run my own startup, Linklet, a campus marketplace for Kenyan university students.

My background combines software development with business and education. I am pursuing a Bachelor of Education (Arts), combining Business Studies and Geography, at Karatina University, and I completed a 12-week Teaching Practice placement as a Geography and Business Studies teacher. This gives me a strong grounding in structured communication, planning and understanding how people learn and use products.

I build with modern AI coding tools (including Claude and OpenCode) alongside my own design, testing and debugging, and I am transparent about this in how I describe my work.

Core Capabilities

- **Web development:** responsive web applications, real-time features, search and filtering, admin dashboards and payment integration.
- **AI development:** integrating language models into workflows and building intelligent, privacy-conscious assistants.
- **Workflow automation:** connecting data sources, AI analysis and email delivery into scheduled, automated reports.
- **Custom software:** purpose-built desktop and web utilities designed around specific user needs.
- **Product thinking:** identifying real problems, defining scope, shipping in phases and iterating on feedback.

Featured Projects

1. Linklet: Campus Marketplace

Category: Web application · **Status:** Beta · **Role:** Founder and sole developer

Linklet helps students and the surrounding community discover goods and connect buyers with sellers. It is a marketplace for goods, not a gig or freelance platform.

Problem. Students buy and sell items informally through scattered social media groups, which makes discovery slow and trust difficult.

Solution. A dedicated marketplace with structured listings, search and filtering, verified sellers and direct buyer-seller messaging.

Key features

- Multiple listing types and listing management.
- Search and filter system.
- Seller verification and an administrator panel.
- Real-time chat using Socket.io.
- M-Pesa (Daraja STK Push) payment integration.
- Dark mode and mobile-friendly interface.

Technology. React and Tailwind CSS frontend deployed on Vercel; Node.js and Express backend; PostgreSQL database; Cloudinary and local upload handling for images; a custom .co.ke domain with professional email.

Engineering challenges and decisions

- Diagnosed and fixed a critical administrator lockout caused by a race condition between maintenance-mode activation and authentication context loading.
- Worked through free-tier hosting limits, including backend cold starts and dropped WebSocket connections, after evaluating several hosting options.
- Replaced a failing unsigned image-upload approach with a reliable local upload fallback.

Future direction. Additional listing categories and business advertising (planned, not yet implemented).

2. FilePilot: Windows File Intelligence

Category: Desktop application · **Status:** Version 0.1.0 pre-release, in real-world testing

FilePilot began with a practical problem: my own C drive was almost full and I could not tell what was consuming the space. It has grown into an integrated Windows application that helps people understand, clean and organise their storage safely.

Built features (in testing)

- Storage analysis with an expandable folder tree, largest-first sorting, size bars, cancellable scans and a free-space report.
- Safe deletion routed through a central safety layer.
- Smart Clean, a file organiser and a duplicate finder with CSV export and exclusions.
- Security Centre with Microsoft Defender integration.
- Explorer diagnostics and a recovery assistant.
- Accessibility and upgrade tests, plus packaging scripts for an installer.

Technology. Python, PySide6 (Qt) desktop interface and SQLite.

Current status. All planned feature phases are built. Remaining work is wider Windows testing, an installer build and an accessibility audit. A course-wise Smart Organiser for students is a planned future feature.

3. NSE Intelligence Agent: Automated Market Briefings

Category: Automation and AI · **Status:** Built and working as a prototype

An automated workflow that collects Kenyan market information, runs it through an AI analysis step and emails a formatted briefing.

How it works

1. A scheduled trigger runs on weekdays at 4:30 PM East Africa Time.
2. HTTP nodes pull price data for blue-chip stocks (Safaricom, Equity and KCB) and headlines from a news RSS feed.
3. An AI analysis step (Claude) summarises the information.
4. The report is formatted and delivered by email over SMTP.

Technology. n8n workflow automation, REST/HTTP requests, RSS, Claude and SMTP.

Important note. The output is automated analysis for information only. It is not financial advice and does not predict share prices.

4. MICA: Local AI Assistant for Windows

Category: AI and desktop software · **Status:** In development (product vision; capabilities to be documented)

MICA is intended to be a local, privacy-oriented AI assistant for Windows with conversational interaction, an animated desktop avatar, file and document workflows, research support and controlled computer interaction, with user approval for consequential actions. A detailed case study will be added once current capabilities are documented.

Additional Projects

- **Smart Sales Assistant:** a WhatsApp automation platform for businesses, with a React and Recharts admin dashboard and a Flask (Python) backend. *In development.*
- **WordCraft:** a writing and proofreading add-in with an in-browser document editor. *In development.*
- **AI Telegram Bot and UniLink:** earlier prototypes.

Technical Skills (linked to projects)

- **Frontend:** JavaScript, React, Tailwind CSS, Recharts (Linklet, Smart Sales Assistant).
- **Backend and APIs:** Node.js, Express, Flask, REST APIs, Socket.io (Linklet, Smart Sales Assistant).
- **Databases:** PostgreSQL (Linklet), SQLite (FilePilot).
- **Desktop:** Python and PySide6 (FilePilot).
- **Automation and AI:** n8n, HTTP and RSS integration, SMTP email, language-model integration (NSE Intelligence Agent).
- **Payments and media:** M-Pesa Daraja STK Push, Cloudinary (Linklet).
- **Tools and deployment:** Git and GitHub, Vercel, VS Code, AI-assisted development with Claude and OpenCode.

- **Currently strengthening:** Python for data analysis and reporting.

Services

- **Web Development:** websites, web applications, responsive interfaces and integrations.
- **AI Development:** AI-powered applications, assistants and model integrations for practical workflows.
- **Workflow Automation and Integrations:** automating repetitive tasks, connecting services, cleaning and converting data, and generating scheduled reports.
- **Custom Software Development:** purpose-built web and desktop utilities designed around specific needs.

I work as an independent builder and communicate honestly about scope, timelines and what is realistically achievable.

Education

Bachelor of Education (Arts), Karatina University School of Education and Social Sciences, Department of Curriculum Technology and Management. Arts degree with teaching subjects in Business Studies and Geography. Final year, graduating in November 2027.

Relevant coursework includes Operations Research (decision theory, linear programming, simulation), Quantitative Methods, Business Environment, and Human Growth and Development.

Teaching Experience

Teaching Practice, Mathaithi Mixed Day Senior School, Karatina (12 weeks)

- Taught Grade 10 Geography and Business Studies under Kenya's Competency-Based Curriculum (CBC).
- Produced CBC-aligned exams, marking schemes and a Table of Item Specification, plus lesson plans and classroom resources.
- Wrote Teaching Practice project reports applying established learning and motivation theories.

Professional Development and Current Activities

- Participating in a six-month employability and entrepreneurship mentorship programme for young university educators (started 27 October 2026).
- Applied to the Anthropic Claude Campus Ambassador Program (Builder Club Lead track) and the ECF–YARA AI, Ethics and Climate Governance Fellowship 2026.
- Building FilePilot toward its first public release and developing a professional business presence for freelance work.
- Taking on freelance data and workflow tasks while building a track record.

Goals

- Ship Linklet's next stage and grow its user community in Kenyan campuses.
- Release FilePilot publicly after Windows testing.

- Deliver automation, AI and web solutions for clients who need practical, reliable software.
- Combine technology and education to contribute to Kenya's digital and economic development.

Contact

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